

CO5 – ADVANCED REINFORCEMENT LEARNING ASSESSMENT

Simulation-Based Assessment

STUDENT DETAILS

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Course Code	C05	Course Outcome	C05
Assessment Tool	Simulation-Based Assessment	AT No.	2
Assessment Type	Advanced Reinforcement Learning	Bloom Level	L3 / L4
Faculty	Dr. Vincent Gnanaraj	Marks	_____/____

Assessment Details

CO5 – Advanced Reinforcement Learning Assessment

Assessment Question – Hierarchical RL & MAXQ (Apply – L3)

Scenario-Based Assessment – Meta-Learning, Multi-Agent RL & POMDP (Analyse – L4)

Instruction: Each question contains TWO missing blocks. Select the option pair that correctly completes BOTH blanks.

Assessment Question – Hierarchical RL & MAXQ (Apply – L3)

1. A warehouse robot uses a high-level policy to choose a task and a low-level policy to execute actions. HRL therefore separates ____ decisions from ____ decisions.

A. local / global

B. strategic / operational

C. random / statistical

D. reward / state

2. A high-level controller selects NavigateRoom, while a lower controller performs wheel movements. The first controller works at a ____ time scale and the second at a ____ time scale.

A. slower / faster

B. faster / slower

C. local / global

D. fixed / random

3. A robot learns NavigateToDoor once and reuses it for several missions. This demonstrates skill ____ and can improve learning ____.

A. duplication / latency

B. reuse / efficiency

C. encoding / noise

D. removal / variance

4. A long-horizon task has rewards only at the end. Splitting it into reusable subtasks can reduce the effective ____ horizon and improve ____.

A. reward / encryption

B. state / compression

C. policy / rendering

D. planning / exploration

5. A high-level policy chooses a sub-goal and waits until the skill finishes before selecting another. This uses temporal ____ and explicit skill ____.

A. randomization / encoding

B. abstraction / termination

C. normalization / sampling

D. compression / clipping

6. If an HRL controller changes sub-goals before a skill completes, the hierarchy may suffer from poor ____ design and unstable ____ control.

A. reward / visual

B. dataset / statistical

C. state / network

D. termination / temporal

7. A MAXQ hierarchy represents a mission using parent and child tasks. The main value function is therefore ____ into smaller task-level ____.

- A. normalized / episodes
- B. encrypted / states
- C. decomposed / values
- D. duplicated / datasets

8. In MAXQ, a child task may obtain a high local value but still reduce the parent task's outcome. Local optimization does not always imply global ____ and task-level ____.

- A. accuracy / encryption
- B. optimality / coordination
- C. variance / sampling
- D. randomness / compression

9. A parent task invokes Search, Pick, and Deliver as child tasks. This hierarchy supports task ____ and reusable ____.

- A. sampling / rewards
- B. compression / states
- C. encryption / gradients
- D. composition / subtasks

10. A designer encodes Scan → Identify → Pick → Place before learning. This reduces the exploration ____ but may reduce policy ____.

- A. space / flexibility
- B. dataset / variance
- C. reward / accuracy
- D. state / latency

11. A skill can be used by several parent tasks when its goal conditions are compatible. This property is called skill ____ and supports policy ____.

- A. reuse / modularity
- B. sampling / encryption
- C. duplication / latency
- D. clipping / variance

12. A subtask takes different numbers of primitive actions in different states. Its value must account for variable ____ and accumulated ____.

- A. width / memory
- B. depth / pixels
- C. frequency / noise
- D. duration / reward

13. An over-constrained hierarchy prevents the agent from discovering a shorter route. The structure improves guidance but reduces behavioral ____ and task ____.

A. flexibility / adaptability

B. variance / compression

C. latency / accuracy

D. reward / sampling

14. A high-level policy chooses Explore, and the lower policy decides movement. The high-level action is an abstract ____ while movement is a primitive ____.

A. reward / state

B. state / skill

C. skill / action

D. gradient / episode

15. A MAXQ parent combines child-task values with the effects of completing those children. This represents hierarchical ____ rather than a single flat ____.

A. encryption / reward

B. compression / episode

C. randomization / dataset

D. decomposition / policy

16. A meta-RL agent trains on many traffic patterns and adapts quickly to a new pattern. It learns a transferable ____ rather than one fixed ____.

A. gradient / episode

B. dataset / reward

C. state / action

D. strategy / policy

17. A meta-trained agent adapts poorly to unrelated tasks. This indicates that meta-learning depends on the ____ between tasks and may fail under distribution ____.

A. relatedness / shift

B. reward / clipping

C. policy / compression

D. state / encoding

18. A meta-RL system learns an initialization that changes rapidly using few new samples. This improves adaptation ____ and reduces data ____.

A. speed / requirements

B. state / compression

C. variance / pixels

D. reward / latency

19. Two vehicles update their policies simultaneously. One vehicle's changing behavior alters the other's learning target. The effective environment becomes ____ and learning becomes ____.

A. non-stationary / coupled

B. observable / fixed

C. deterministic / independent

D. stationary / isolated

20. Four agents receive one delayed team reward. They cannot identify individual contributions. This creates a ____ assignment problem and makes credit ____ difficult.

A. reward / clipping

B. credit / assignment

C. state / transition

D. policy / sharing

21. Two agents must choose complementary actions to succeed. Independent learning may fail because success depends on joint ____ and coordinated ____.

A. states / rewards

B. actions / policies

C. episodes / observations

D. datasets / gradients

22. During training, a MARL system sees all agents' observations, but during deployment each agent sees only local data. This is centralized ____ with decentralized ____.

A. reward / learning

B. sampling / planning

C. execution / training

D. training / execution

23. In competitive MARL, one agent changes its policy and alters the strategies faced by others. This creates strategic ____ and reduces environmental ____.

A. sampling / accuracy

B. coupling / stationarity

C. reward / latency

D. compression / variance

24. A robot cannot observe a hidden obstacle directly and receives noisy sensor readings. The problem is partially ____, so the agent maintains a ____ state.

A. deterministic / policy

B. observable / belief

C. stationary / reward

D. episodic / gradient

25. A POMDP agent combines current evidence with previous observations before acting. Its decision uses a ____ state rather than only the current ____.

- A. reward / state
- B. gradient / episode
- C. policy / action
- D. belief / observation

26. The same sensor reading can lead to different actions because the observation history differs. Under partial observability, the agent therefore uses ____ information to infer the hidden ____.

- A. random / reward
- B. visual / gradient
- C. historical / state
- D. fixed / policy

27. An RL controller reduces waiting time by violating an unstated safety rule. The reward is ____, causing reward ____.

- A. normalized / clipping
- B. dense / sampling
- C. misspecified / hacking
- D. sparse / shaping

28. An RL system performs well in simulation but fails with real traffic. The problem reflects simulation-to-real ____ and requires stronger ____ testing.

- A. reward / rendering
- B. compression / encryption
- C. sampling / clipping
- D. shift / validation

29. A safety-critical system combines HRL, POMDP reasoning, and MARL. Hierarchy adds structural complexity, POMDP adds ____, and MARL adds agent ____.

- A. compression / latency
- B. normalization / variance
- C. sampling / encryption
- D. uncertainty / interaction